

Mock Test — The Amazing World of Solutes, Solvents, and Solutions

A mixed paper covering the whole chapter. Total: 20 marks. Section A is interactive.

Objective (1 mark each)

Q1 • MCQ

[1]

In a salt solution, water is the —

- (a) solute (b) solvent (c) solution (d) mixture

Q2 • MCQ

[1]

A solution that can dissolve no more solute at a given temperature is —

- (a) unsaturated (b) saturated (c) dilute (d) pure

Q3 • MCQ

[1]

Solubility of most solids _____ with temperature.

- (a) increases (b) decreases (c) is unchanged (d) becomes zero

Q4 • MCQ

[1]

Solubility of gases _____ as temperature rises.

- (a) increases (b) decreases (c) stays same (d) doubles

Q5 • MCQ

[1]

The dissolved gas vital for fish is —

- (a) oxygen (b) nitrogen (c) hydrogen (d) argon

Q6 • TRUE / FALSE

[1]

Sand in water forms a solution.

- (a) True (b) False

Q7 · FILL IN THE BLANK

[1]

The maximum solute a solvent can dissolve is its _____.

Q8 · FILL IN THE BLANK

[1]

A solution with little solute is called _____.

Short answer (2 marks each)

Q9

[2]

Define solute and solvent with an example.

Q10

[2]

How does temperature affect the solubility of solids and of gases?

Q11

[2]

What is a saturated solution? How can it dissolve more solute?

Long / application (3 marks each)

Q12

[3]

Describe how you would show that hot water dissolves more of a solid than cold water.

Q13

[3]

Explain why cold water supports more aquatic life and why a warm fizzy drink goes flat, using gas solubility.

[Check yourself](#) → [Answer key](#)